

Bit Manipulations / Bit Masking

Number of Set Bits / *Hamming Weight*

0110	n = 6	o/p	2
1000	n = 8	o/p	1
0111	n = 7	o/p	3
0101	n = 5	o/p	2
0000	n = 0	o/p	0

n = 7
 0111 & 0001 = 1 → True
 n >> 1 = 011 & 001 = 1
 n >> 1 = 01 & 001 = 1
 n >> 1 = 0 & 1 = 0

count = 0
 ++ 1
 ++ 2
 ++ 3

→ b4 32 16 8 4 2 1 ← Acceptable
 16 → 1 0 0 0 0 1
 MSB 5 4 3 2 1 LSB

2 (20) → 1 0 1 0 0 → -1
 0 0 0 0 → -1

[n <= 0] (-1) (1)
 7 - 0111 → -1

Power of 2

[-ve ... 0] invalid

[3, 5, 7, 9, 11, 12, ... Unacceptable]
 1, 2, 4, 8, 16, 32, ... OK

(N & (N-1)) != 0
 7 (7-1) 6 → 0110
 0111

7 & 6
 0111
 0110
 0110 ✓

8 & 7
 1000
 0111
 0000 ✓

Base Case power of 2
 8 4 2 1
 1 0 0 0
 0 1 1 1
 (N <= 0 || N & (N-1) != 0)
 return -1;

int position = 1
 while ((N & 1) == 0) {
 N = N >> 1;
 position++;
 }
 while (1 & 1 == 0)

1000
 4321
 1000
 100
 10
 1

(n = 1) & 1 == 0
 return position;

* Total Number of Bits →

n = 5

1 → 5
 1 0001 → 1
 2 0010 → 1
 3 0011 → 2
 4 0100 → 1
 5 0101 → 2
 (7)

Minimum Absolute Difference

[4, 2, 1, 3] → ascending order

4 2 1 3
 i = 1, i < n, i++
 a[i] - a[i-1]
 5 8 3 9
 (i-1) i
 3 5 8 9
 a b
 3-9 = 6
 5-9 = 4
 3-8 = 5
 (3) (5) (8) (9)
 a b
 (1) (2) (3) (4)
 a b
 {a[i-1], a[i]}